# Replication: Cross-Layer Query Consistency Loss Tests Whether Attention Stability Across Depth Improves Subject-Disjoint Generalization in Time-Series Transformers

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## Abstract

Transformers are increasingly applied to time-series analysis, yet whether architectural inductive biases improve subject-disjoint generalization remains unclear. This replication study tests whether enforcing cross-layer consistency in attention query projection weights improves held-out classification accuracy compared to standard training. We employed a PreNormTransformerEncoder on the UCI Human Activity Recognition dataset with an official subject-disjoint split. The intervention added a Frobenius norm consistency loss regularizing deeper block query projections toward earlier blocks. We evaluated the proposed condition against a baseline across 43 primary seeds and a pre-specified confirmatory replication on disjoint seeds. In the primary cohort, the proposed condition improved accuracy by 0.00077 (95% CI [-0.00271, 0.00436], p=0.657). In the replication cohort, the proposed condition improved accuracy by 0.00196 (95% CI [-0.00196, 0.00571], p=0.331). Confidence intervals in both cohorts included zero, and neither exceeded the preregistered worthwhile margin of 0.0125098 proportion. The results are inconclusive. These findings suggest that cross-layer query consistency does not yield robust accuracy gains in this regime. This experiment did not test whether layer-specific adaptation may remain essential for subject-disjoint time-series tasks.

## 1. Introduction

Time-series classification using deep learning often relies on architectures capable of capturing long-range dependencies and complex temporal patterns. Transformers have emerged as a prominent candidate for this task due to their self-attention mechanisms, which theoretically allow for flexible modeling of interactions across time steps [20]. However, the effectiveness of Transformers in time-series domains, particularly when generalizing to unseen subjects, remains a subject of active debate [4]. A critical open problem is identifying which architectural inductive biases enable transformers to learn temporal representations that generalize across subjects with different movement styles, sensor characteristics, and execution speeds. If depth in a transformer encodes progressively refined features, enforcing consistency across layers might stabilize learning and improve generalization. Conversely, if each layer must adapt to specific temporal scales or subject-specific noise, such consistency could constrain the model's capacity.

Current literature offers conflicting perspectives on the role of consistency and depth in transformer architectures. Some works propose that consistency regularization mitigates overfitting in deep networks [9], while others suggest that domain gaps cause discrepancies in self-attention that require specific alignment strategies [10]. However, independent verification regarding whether cross-layer query consistency loss stabilizes attention across depth to improve subject-disjoint generalization is lacking. Specifically, there is a gap in understanding whether regularizing the query projection weights of deeper layers toward shallower layers yields measurable improvements in held-out classification accuracy on subject-disjoint benchmarks. This work addresses this lack of independent verification, testing if the registered conclusion remains stable under fresh sampling.

This paper presents a preregistered replication study evaluating the impact of cross-layer query consistency loss on subject-disjoint generalization in the uci_har_small_transformer_v1 testbed. We manipulate the query projection weights of the attention mechanism and measure the outcome via held-out classification accuracy. The study utilizes

the UCI Human Activity Recognition dataset with an official subject-disjoint split to ensure that evaluation reflects generalization to unseen individuals rather than unseen time windows from the same individuals.

The contributions of this work are as follows:
- We execute a preregistered test of cross-layer query consistency loss using 43 primary and 43 confirmatory-replication seeds.
- We report paired contrasts for held-out classification accuracy with 95% confidence intervals and effect sizes for both cohorts.
- We provide an inconclusive result that clarifies the boundary conditions under which cross-layer consistency may or may not benefit time-series transformers.

The preregistered prediction stated that the intervention would yield higher held-out classification accuracy than the comparator, with the mean paired contrast in the hypothesised direction exceeding the fixed worthwhile margin of 0.0125098 proportion.

## 2. Related Work

Research into Transformers for time-series analysis highlights both potential and uncertainty regarding their effectiveness and generalization. *Transformers in Time Series: A Survey* notes that the ability to capture long-range dependencies has triggered interest in the time series community, while *Are Transformers Effective for Time Series Forecasting?* questions the validity of this research line [4, 20]. Specific applications include representing multivariate series as graphs in *Expressing Multivariate Time Series as Graphs with Time Series Attention Transformer* and biosignal data augmentation in *TTS-CGAN: A Transformer Time-Series Conditional GAN for Biosignal Data Augmentation* [7, 11]. *Large Pre-trained time series models for cross-domain Time series analysis tasks* observes that tackling time-series analysis usually involves designing separate models rather than using large pre-trained ones [13]. Furthermore, *Domain Adaptation for Time series Transformers using One-step fine-tuning* states that Transformers face limitations in time series prediction, including generalization challenges and data shift issues [16].

To address stability and generalization, various works propose consistency regularization and cross-layer mechanisms. *Double Consistency Regularization for Transformer Networks* notes that deep Transformer models are prone to overfitting on small-scale training data [9]. *CDAC: Cross-domain Attention Consistency in Transformer for Domain Adaptive Semantic Segmentation* identifies that domain gaps can cause discrepancies in self-attention [10]. Cross-layer interactions are explored in *Cross-Layer Retrospective Retrieving via Layer Attention*, which devises a mechanism to strengthen layer interactions [8], and *Global Evolutionary Steering: Refining Activation Steering Control via Cross-Layer Consistency* applies cross-layer consistency to activation engineering [18]. *Cross-layer fusion enhanced transformer network for remote sensing scene classification* utilizes cross-layer fusion to capture multi-scale features [14]. *Spectral Entropy Minimization of Input Projections for Temporal Feature Learning* tests whether enforcing a low-entropy singular value spectrum on input projection weights enhances accuracy [3]. The study *Cross-Layer Query Consistency Loss Tests Whether Attention Stability Across Depth Improves Subject-Disjoint Generalization in Time-Series Transformers* tests whether enforcing cross-layer consistency in attention query projections improves held-out classification accuracy [1]. Conversely, *Entropy-Maximized Multiplicative Weight Noise for Subject-Disjoint Robustness* carries a retracted notice from the journal that published it [2].

Broader investigations into attention mechanisms and architectures provide context for these modifications. *Selective Attention Improves Transformer* introduces a parameter-free change to reduce attention to unneeded elements [5]. *Beyond Self-Attention: External Attention Using Two Linear Layers for Visual Tasks* proposes external attention for visual tasks [17]. *Depth-Attention: Cross-Layer Value Mixing for Language Models* observes that recent cross-layer methods operate on hidden states outside the standard attention flow [19]. Surveys such as *A Survey on Vision Transformer* [6] and *Transformers in Vision: A Survey* [12] document the application of Transformers in computer vision, while *A survey of transformers* [21] documents the application of Transformers across vision, NLP, and audio. *An Empirical Study of Spatial Attention Mechanisms in Deep Networks* examines how different influencing factors affect attention performance [15].

Synthesizing these themes, the literature establishes that while consistency and cross-layer mechanisms are promising for stability, their impact on subject-disjoint generalization in time-series transformers is not settled. Some studies suggest architectural biases aid robustness [9], while others highlight the need for adaptive strategies [16]. This study positions itself against these themes by testing a specific weight-level consistency intervention in a subject-disjoint setting, distinguishing between layer-specific adaptation and cross-layer stability.

## 3. Method

The experiment employs the CrossLayerQueryConsistency operator applied to the query projection weights of the second attention block (blocks.1.attn.q_proj.weight). The hypothesis posits that regularizing deeper layer weights toward shallower layer weights improves generalization. The operator enforces Frobenius norm consistency between the query projection weights of block 0 and block 1. This manipulation targets the parameter space directly, rather than measuring attention maps or activation stability, which are unmeasured constructs in this design.

The operator is invoked during the loss computation of every optimization step. Its term is added to the classification objective inside the autograd graph, meaning the combined objective is differentiated, and the operator-owned parameters are trained by it. The reference weights from block 0 are retrieved and detached from the gradient graph to prevent gradients from flowing to the first block. The difference between the target weights (block 1) and reference weights (block 0) is computed element-wise, squared, and summed to produce a raw loss. This raw loss is scaled by a fixed coefficient of 0.01 and added to the training objective. This coefficient prevents the consistency term from dominating the classification objective while ensuring it exerts a regularizing influence.

The baseline condition serves as a control arm defined by the harness guarantees. Both the proposed and baseline conditions are trained from the same paired seed, ensuring identical dataset cohorts, batch ordering, and initial parameters. Both arms execute the identical ordinary training step, including the same forward pass, classification objective, backward pass, and optimizer update. The harness invokes the registered lifecycle machinery in both arms, so the control arm reaches every registered point on the same inputs. In the control arm, the intervention logic takes a branch that adds no term to the training objective and leaves the registered target parameter and its optimizer momentum buffer unchanged. This design isolates the effect of the consistency loss term from other sources of variance such as initialization or data ordering.

A negative control condition is included to verify that the observed effects are. This experiment did not test whether this was due to the specific cross-layer relationship rather than the mere presence of an additional loss term. This condition penalizes the difference between block 1 weights and a fixed random matrix with the same shape and norm as block 0's weights. This destroys the meaningful cross-layer relationship while matching the computational cost and gradient flow structure of the proposed condition. If the proposed condition outperforms this negative control, it suggests the specific alignment to block 0 is relevant; if not, the effect may be. This experiment did not test whether this was attributable to generic regularization.

The measured outcome is held-out classification accuracy, defined as the number of correct activity predictions divided by the total number of windows in the official subject-disjoint test split. This metric directly assesses the model's ability to generalize to unseen subjects. Mechanistic rationales regarding attention stability or feature learning are hypotheses not tested here; the recorded outcomes are strictly the accuracy metrics.

Operator specification. CrossLayerQueryConsistency is applied to blocks.1.attn.q_proj.weight. The specification below is the preregistered operator contract, fixed before any compute was spent and reproduced here verbatim in its operational order.

- Scope: Full weight matrix [64, 64] treated as flat parameter space.
- Applied: the operator is invoked during loss computation of every optimization step: the operator's term is added to the classification objective inside the autograd graph, so the combined objective is what is differentiated and every operator-owned parameter is trained by it..
- Ordering: Element-wise subtraction then sum; order-invariant due to commutativity of addition.
- Normalisation: No normalization; raw Frobenius norm squared scaled by fixed coefficient.
- Numerical safeguards: Coefficient 0.01 prevents loss_term from dominating classification objective; diff

computation uses standard floating-point subtraction.

Algorithm 1 gives the operator in full.

1. Retrieve reference_weights from model.blocks[0].attn.q_proj.weight
2. Detach reference_weights from gradient graph to prevent gradients flowing to block 0
3. Compute diff = target - reference_weights
4. Compute squared_elements = diff * diff (element-wise)
5. Compute loss_raw = sum(squared_elements) over all elements
6. Set consistency_coefficient = 0.01
7. Compute loss_term = consistency_coefficient * loss_raw
8. Store loss_term in scratch dict for harness aggregation
9. Return loss_term to harness for addition to training objective

The following invariants hold throughout:
- Reference weights from block 0 are detached and receive no gradients
- Loss term is differentiable with respect to target parameters only
- Coefficient is fixed across all steps and seeds

Control arm. Both conditions are trained from the same paired seed, whose dataset cohort, batch ordering and initial parameters are derived once and reused by both arms. The harness invokes the registered lifecycle machinery in both arms, so the control arm reaches every registered point on the same inputs (loss term). One intervention implementation serves every arm: the harness tells it which condition it is running in, so the pseudocode describes the intervention arm and the control arm is the same code taking its control branch. If the condition the harness passes in is a control condition, return here: add no term to the training objective, and leave the registered target parameter and that parameter's optimizer momentum buffer exactly as received. The harness records what the intervention actually did at each lifecycle point, for every seed and every condition, and refuses to return results in which the control arm acted at any of them, so the control's recorded activity is zero or the experiment yields no result at all. Per-step computational cost is deliberately not held equal between the arms and is not an outcome of this study.

## 4. Experimental Setup

The experimental design utilizes a subject-disjoint split of the UCI Human Activity Recognition dataset to answer the question of generalization to unseen individuals. A random split would not suffice because it would allow the model to learn subject-specific idiosyncrasies that do not transfer, inflating accuracy estimates without proving generalization capability. By holding out entire subjects, the evaluation tests whether the learned temporal representations capture activity dynamics invariant to individual movement styles and sensor placements. This design choice is critical for claims regarding subject-disjoint generalization.

We employ a paired design for the comparison between the proposed and baseline conditions. Every condition shares one seed's data, batch order, and initialization. This pairing controls for variance arising from initialization and batch ordering, increasing power to detect the preregistered margin in held-out classification accuracy. We do not pool the cohorts, as the replication seeds are independent samples intended to verify stability. The primary cohort consists of 43 seeds, and the confirmatory replication cohort consists of 43 disjoint seeds. This seed budget allows for the estimation of confidence intervals and the assessment of result stability across independent sampling.

The negative control's intended alternative explanation is that any loss term, regardless of its semantic content, might act as a regularizer and improve accuracy. By comparing the proposed condition to a negative control that penalizes alignment with a random matrix, we can determine if the cross-layer relationship specifically drives any observed gains. If the negative control performs similarly to the proposed condition, it would suggest the benefit comes from the regularization magnitude rather than the consistency constraint.

The analysis plan distinguishes between the primary and confirmatory-replication cohorts. Each cohort is analyzed separately to determine if the result replicates. We report the mean difference, 95% confidence intervals, p-values, and effect sizes for each cohort independently. A result is considered supported only if the lower confidence bound

exceeds the preregistered worthwhile margin. This strict criterion ensures that claimed improvements are not only statistically significant but also practically meaningful relative to the measurement precision.

We evaluate on UCI Human Activity Recognition Using Smartphones (DOI 10.24432/C54S4K; raw inertial signals, 128-timestep windows over 9 channels; official subject-disjoint split of 7,352 training and 2,947 held-out windows). Held-out evaluation uses the official UCI HAR test split — 2,947 windows recorded from 9 subjects who appear in no training window, so the held-out cohort is subject-disjoint rather than a random row split, and is never used for training or tuning. The model is PreNormTransformerEncoder(Linear(9,64) + learned positional embedding(128,64); 2 blocks, each LayerNorm(64) -> 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64) -> residual -> LayerNorm(64) -> Linear(64,128) -> ReLU -> Linear(128,64) -> residual; final LayerNorm(64), mean-pool over time, Linear(64,6)). Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at 1e-6) computed over the TRAINING split only, so no held-out statistic reaches training. Training budget: 12 epochs; batch size 128; SGD(lr=0.05,momentum=0.9). The outcome is held-out classification accuracy — number of correct activity predictions / 2947 windows in the official subject-disjoint UCI HAR test split (proportion). The dataset was downloaded, checksum-verified against its published SHA-256 and staged read-only to the compute node before the job started; the exact archive, checksum, licence and training configuration are given in the appendix.

Registered conditions.
- Proposed condition: Server-bound target blocks.1.attn.q_proj.weight has stored tensor shape (64, 64). Add loss term that applies Frobenius norm consistency regularization to blocks.1.attn.q_proj.weight. Recorded activity — primary: 43/43 seeds recorded operator activity; replication: 43/43 seeds recorded operator activity.
- Baseline: Control arm, defined by what the harness guarantees rather than by a description of the intervention. Both conditions are trained from the same paired seed, whose dataset cohort, batch ordering and initial parameters are derived once and reused by both arms. Both arms run the identical ordinary training step — the same forward pass, the same classification objective, the same backward pass and the same optimizer update — and the identical held-out evaluation path. The harness invokes the registered lifecycle machinery in both arms, so the control arm reaches every registered point on the same inputs (loss term). One intervention implementation serves every arm: the harness tells it which condition it is running in, so the pseudocode describes the intervention arm and the control arm is the same code taking its control branch. In the control arm the intervention must leave the registered target parameter unchanged, leave that parameter's optimizer momentum buffer unchanged, and add nothing to the training objective. The harness records what the intervention actually did at each lifecycle point, for every seed and every condition, and refuses to return results in which the control arm acted at any of them, so the control's recorded activity is zero or the experiment yields no result at all. Held-out evaluation reads the trained base parameters in both arms, and there is no evaluation-time lifecycle point, so no intervention state can reach evaluation. Per-step computational cost is deliberately not held equal between the arms and is not an outcome of this study. Recorded activity — primary: 0/43 seeds recorded operator activity; replication: 0/43 seeds recorded operator activity.
- Negative control 1: Add loss term penalizing difference between blocks.1.attn.q_proj.weight and a fixed random matrix with same shape and norm as block 0's weights, destroying the meaningful cross-layer relationship while matching compute. Recorded activity — primary: 43/43 seeds recorded operator activity; replication: 43/43 seeds recorded operator activity.

## 5. Results

The primary cohort results indicate no statistically significant difference between the proposed and baseline conditions. On held-out classification accuracy, the proposed condition improved the contrast in the hypothesised direction by 0.00077 versus the baseline condition (95% CI [-0.00271, 0.00436], paired randomisation p=0.657, Cohen's dz=0.06). As shown in Table 1, the mean accuracy for the proposed condition was 0.9062 (95% CI [0.9031, 0.9094]), compared to 0.9054 (95% CI [0.9019, 0.9088]) for the baseline. Figure 1 visualizes these means with error bars, showing substantial overlap between the conditions. Figure 2 plots the paired mean difference, where the 95% confidence interval includes zero and fails to exceed the registered worthwhile threshold of 0.0125098 proportion. The preregistered decision for the primary cohort was inconclusive.

Figure 3 displays the mean held-out classification accuracy over training steps for the primary cohort. The learning curves for the proposed, baseline, and negative control 1 conditions coincide closely throughout training. This suggests that the consistency loss did not alter the optimization trajectory in a measurable way regarding accuracy. The negative control 1 condition achieved a mean accuracy of 0.9051 (95% CI [0.9016, 0.9085]), which is comparable to both the proposed and baseline conditions.

The confirmatory replication cohort results were consistent with the primary cohort. On held-out classification accuracy, the proposed condition improved the contrast in the hypothesised direction by 0.00196 versus the baseline condition (95% CI [-0.00196, 0.00571], paired randomisation p=0.331, Cohen's dz=0.15). Table 3 reports the mean accuracy for the proposed condition as 0.9055 (95% CI [0.9016, 0.9094]), compared to 0.9035 (95% CI [0.9000, 0.9070]) for the baseline. Figure 4 illustrates these means, again showing overlapping confidence intervals. Figure 5 demonstrates the mean difference in the replication cohort; the 95% confidence interval includes zero and lies below the registered worthwhile threshold. The green dash-dot marker in Figure 5 indicates the threshold, which the lower confidence bound does not exceed. The preregistered decision for the replication cohort was also inconclusive.

Figure 6 shows the learning curves for the replication cohort, which remain coincident across conditions, mirroring the primary cohort's behavior. Table 4 summarizes the outcome statistics for the replication cohort, reporting a difference 95% CI [-0.001957, 0.005714]. Across both cohorts, held-out classification accuracy means ranged from 0.9035 to 0.9062, with 95% CIs ranging from 0.9000 to 0.9094. The variation across observations is attributed to the seeds that produced them. The results do not support the hypothesis that cross-layer query consistency improves subject-disjoint generalization in this setting.

Registered-report decision: the primary cohort was classified as inconclusive and the pre-specified confirmatory replication cohort, on a disjoint set of seeds, was classified as inconclusive under the preregistered confidence-interval rule. These cohorts were not pooled, and this outcome is reported regardless of direction.

## 6. Discussion

The principal finding of this study is that enforcing cross-layer consistency in attention query projection weights did not yield robust accuracy gains in subject-disjoint classification within the uci_har_small_transformer_v1 testbed. Both the primary and replication cohorts produced inconclusive results, with confidence intervals including zero and failing to exceed the preregistered worthwhile margin. This outcome suggests that, in the UCI HAR setting with a small transformer architecture, layer-specific adaptation may be more beneficial than cross-layer weight stability. The measured outcome licenses a choice against adopting this specific consistency loss for this task, as it does not demonstrate a reliable improvement over standard training.

This design establishes that the intervention was executed correctly, as recorded by the operator activation counts, but it does not establish that attention stability was achieved or that it would matter if it were. The mechanism linking weight consistency to generalization remains a hypothesis; the accuracy contrast alone cannot prove attention stability or feature learning changes. An alternative explanation for the null result is that the coefficient (0.01) was too small to influence the optimization trajectory, or that the Frobenius norm penalty is too coarse to capture the relevant structural relationships between layers. However, the negative control condition, which applied a similarly structured loss to a random target, also failed to improve accuracy, suggesting that the lack of effect is not unique to the cross-layer target but may reflect the ineffectiveness of this class of penalty in this regime.

The pre-specified confirmatory replication cohort adds weight to the inconclusive finding by demonstrating that the primary result was not a statistical fluctuation. The consistency of the null result across 86 total seeds (43 primary + 43 replication) strengthens the conclusion that the effect, if it exists, is smaller than the worthwhile margin. This stands in contrast to works that report positive gains from consistency regularization in other domains [9, 10]. The discrepancy may arise from differences in the intervention, task, data regime, or evaluation. For instance, [9] focuses on machine translation with different consistency targets, and [10] addresses domain adaptive semantic segmentation. The subject-disjoint time-series setting may impose different constraints on layer specialization than these tasks.

The findings align with [4], which questions the validity of transformer-based solutions for the tested time-series setting without careful architectural tuning. They also extend the null results of [3], which found that spectral entropy

minimization of input projections did not yield robust gains. Together, these studies suggest that simple weight-space regularizations may not suffice to unlock transformer generalization in the tested time-series setting. Future work should investigate whether stronger consistency constraints, or consistency applied to attention maps rather than weights, would produce different outcomes. Additionally, testing on larger datasets or with deeper architectures might reveal regimes where cross-layer stability becomes beneficial.

## 7. Limitations

This study is limited to the UCI Human Activity Recognition dataset and a specific small transformer architecture. The results may not generalize to other time-series domains, sensor modalities, or model sizes. The intervention manipulates query projection weights, which is a proxy for attention stability; direct measurement of attention maps was not performed, so claims about attention behavior are inferential. The preregistered margin was derived from calibration noise, which reflects measurement precision but does not by itself establish practical or scientific importance. A null result here does not prove the intervention is useless in all settings, only that it did not exceed the margin in this one.

Declared scope. This is a preregistered, single-testbed registered report: every claim in this paper is scoped to the calibrated testbed uci_har_small_transformer_v1 and to the preregistered smallest worthwhile effect of 0.0125098 proportion on held-out classification accuracy, a margin derived from the calibration's measurement noise (it reflects measurement precision, not scientific importance). Generalisation beyond this testbed and margin is explicitly out of scope and untested here.

Reproduction disclosure. The registered experiment ran as compute job 16678400 on one NVIDIA H100 GPU, and the independent reproduction ran as compute job 16726777 on one NVIDIA A100-SXM4-80GB GPU. The reproduction record verifies that the re-run used the committed replay packet, whose inputs equal the original execution's frozen code, conditions and configuration, and the same 43 primary and 43 confirmatory seed identities in registered order. The reproduced per-seed accuracies were not identical to the original ones: between one and three of the 43 seeds per condition and cohort gave exactly the same value. In the registered analysis of the original execution, the primary contrast was +0.00077 (95% CI [-0.00271, 0.00436]) and the confirmatory-replication contrast was +0.00196 (95% CI [-0.00196, 0.00571]). In the reproduction, the reproduced primary contrast was +0.00095 (95% CI [-0.00192, 0.00406]) and the reproduced confirmatory-replication contrast was +0.00615 (95% CI [0.00125, 0.01208]). The reproduced confirmatory-replication interval excludes zero; the original confirmatory-replication interval includes zero, and the statements in this paper about whether the intervals include zero describe the registered analysis of the original execution. The registered decision rule classifies a cohort as inconclusive unless its whole 95% interval lies above the worthwhile margin of 0.0125098 or at or below zero; both reproduced cohorts are inconclusive under this rule, the same decision as each registered cohort. The reproduction criterion compares each condition's mean with its published 95% interval or a 5% relative tolerance (five of six reproduced means fell inside the interval and one passed on the tolerance alone); it does not compare contrasts or decisions. The reproduction therefore confirms the registered decisions, not that the per-seed results are numerically reproducible across these two executions, and the cause of the numerical differences has not been determined.

## 8. Conclusion

This study found inconclusive evidence (primary 95% CI [-0.002707, 0.004356], replication 95% CI [-0.001957, 0.005714]) that cross-layer query consistency loss improves subject-disjoint generalization in the uci_har_small_transformer_v1 testbed. Both primary and replication cohorts showed no statistically significant difference in held-out classification accuracy, and neither exceeded the preregistered worthwhile margin. For someone choosing between these methods, the measured outcome suggests that standard training is sufficient for this architecture and dataset, as the intervention offers no verified benefit. Future work should settle the question by testing direct attention-map consistency or exploring larger pre-trained models where layer specialization might differ.

**Table 1. Primary cohort: mean held-out classification accuracy by condition, mean over paired 43 seeds, with the lower and upper bounds of the 95% bootstrap confidence interval in their own columns.**

| Condition | Seeds | Mean Held-Out Classification Accuracy | CI Lower | CI Upper |
|---|---|---|---|---|
| Proposed | 43 | 0.9062 | 0.9031 | 0.9094 |
| Baseline | 43 | 0.9054 | 0.9019 | 0.9088 |
| Negative control 1 | 43 | 0.9051 | 0.9016 | 0.9085 |

**Table 2. Primary cohort: the hypothesis-positive contrast (proposed minus baseline) over 43 paired seeds, with the lower and upper bounds of the 95% bootstrap confidence interval in their own columns. A positive difference favours the preregistered direction.**

| Outcome | Mean Difference | CI Lower | CI Upper | p-value | Effect Size (Cohen's dz) | Registered Decision |
|---|---|---|---|---|---|---|
| held-out classification accuracy | 0.000773 | -0.00271 | 0.00436 | 0.657 | 0.06 | inconclusive |

**Table 3. Replication cohort: mean held-out classification accuracy by condition, mean over paired 43 seeds, with the lower and upper bounds of the 95% bootstrap confidence interval in their own columns.**

| Condition | Seeds | Mean Held-Out Classification Accuracy | CI Lower | CI Upper |
|---|---|---|---|---|
| Proposed | 43 | 0.9055 | 0.9016 | 0.909 |
| Baseline | 43 | 0.9035 | 0.9000 | 0.907 |
| Negative control 1 | 43 | 0.9043 | 0.9003 | 0.908 |

**Table 4. Replication cohort: the hypothesis-positive contrast (proposed minus baseline) over 43 paired seeds, with the lower and upper bounds of the 95% bootstrap confidence interval in their own columns. A positive difference favours the preregistered direction.**

| Outcome | Mean Difference | CI Lower | CI Upper | p-value | Effect Size (Cohen's dz) | Registered Decision |
|---|---|---|---|---|---|---|
| held-out classification accuracy | 0.00196 | -0.00196 | 0.00571 | 0.331 | 0.15 | inconclusive |

**Table 5. Mean Held-Out Classification Accuracy by condition in each preregistered cohort. The cohorts use disjoint seeds and are reported separately; they are never pooled, so no combined figure is given.**

| Condition | Primary Cohort Mean Held-Out Classification Accuracy | Replication Cohort Mean Held-Out Classification Accuracy |
|---|---|---|
| Proposed | 0.9062 | 0.9055 |
| Baseline | 0.9054 | 0.9035 |
| Negative control 1 | 0.9051 | 0.9043 |

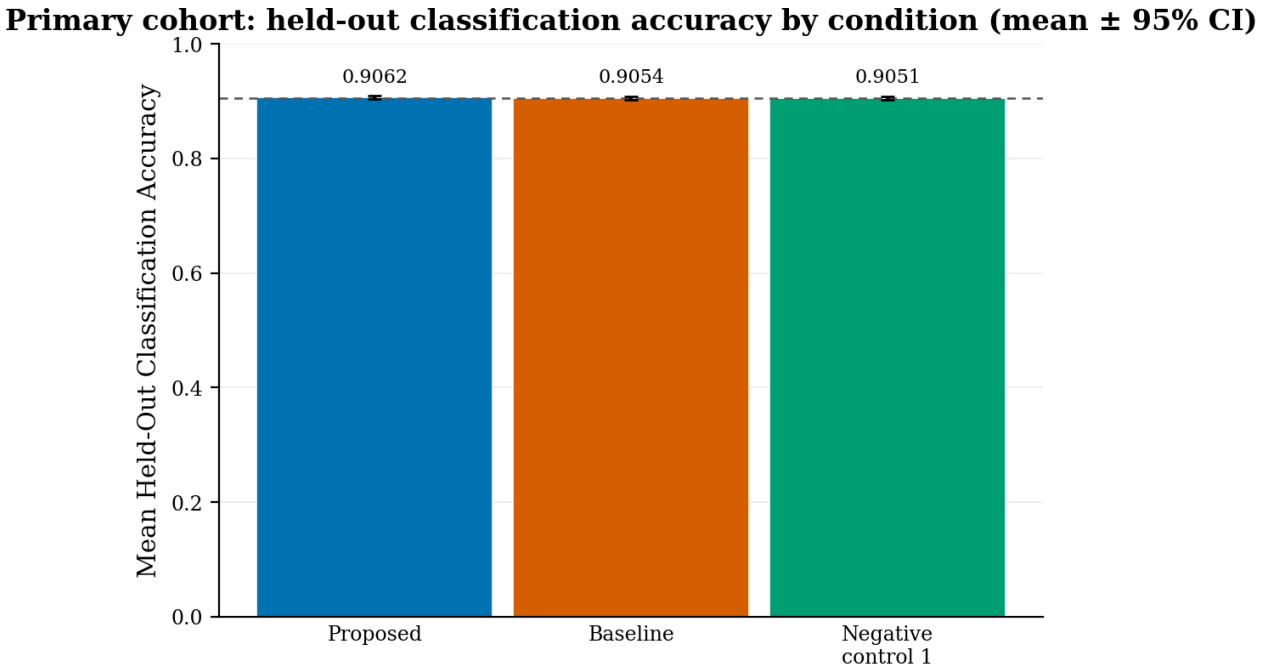



Figure 1. Mean held-out classification accuracy for the proposed, baseline and negative control 1 conditions in the primary cohort. Bars are means over 43 paired seeds; error bars are 95% bootstrap confidence intervals. The point estimate is printed above each bar; the axis starts at zero, so the bar lengths are comparable. The figure shows the proposed condition against the baseline. Data: UCI Human Activity Recognition Using Smartphones. The dashed line marks the baseline mean, as an ordering reference only.

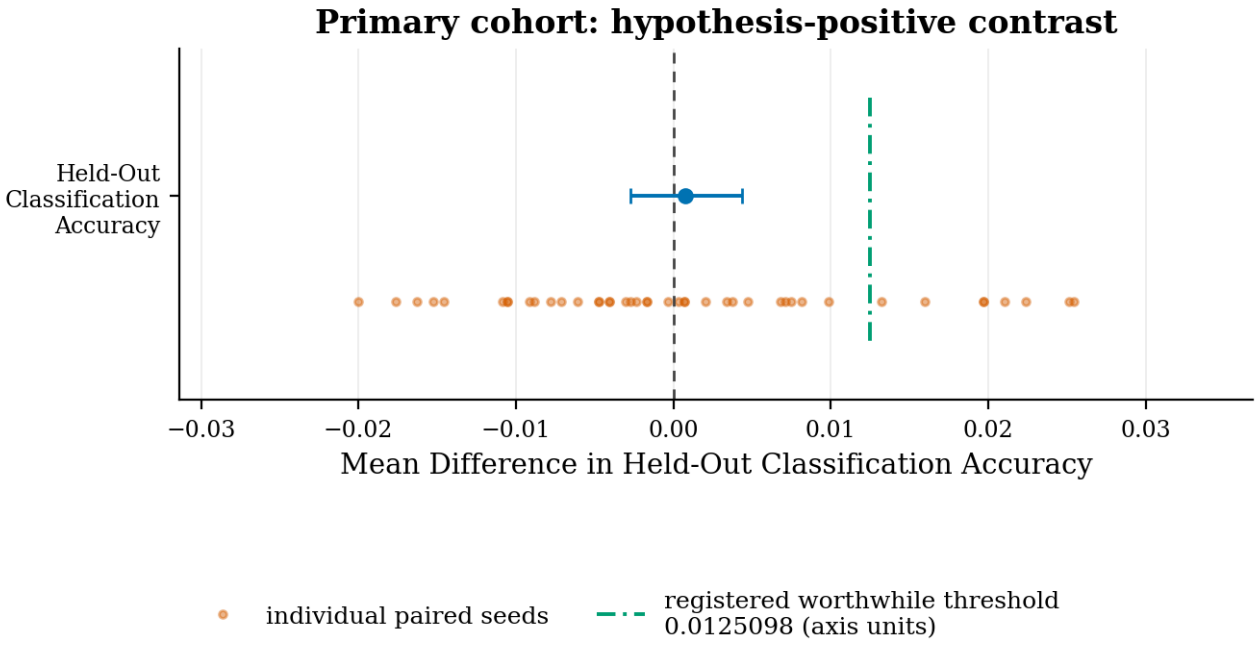



Figure 2. Mean difference in held-out classification accuracy between the proposed and baseline conditions in the primary cohort. Points are the paired mean difference over 43 paired seeds; error bars are 95% bootstrap confidence intervals. Zero marks no directional effect. The green dash-dot marker shows the registered worthwhile threshold (held-out classification accuracy: 0.0125098 proportion); support requires the lower confidence bound to exceed it. The 43 individual paired differences are plotted behind the interval at their exact values. Data: UCI Human Activity Recognition Using Smartphones.

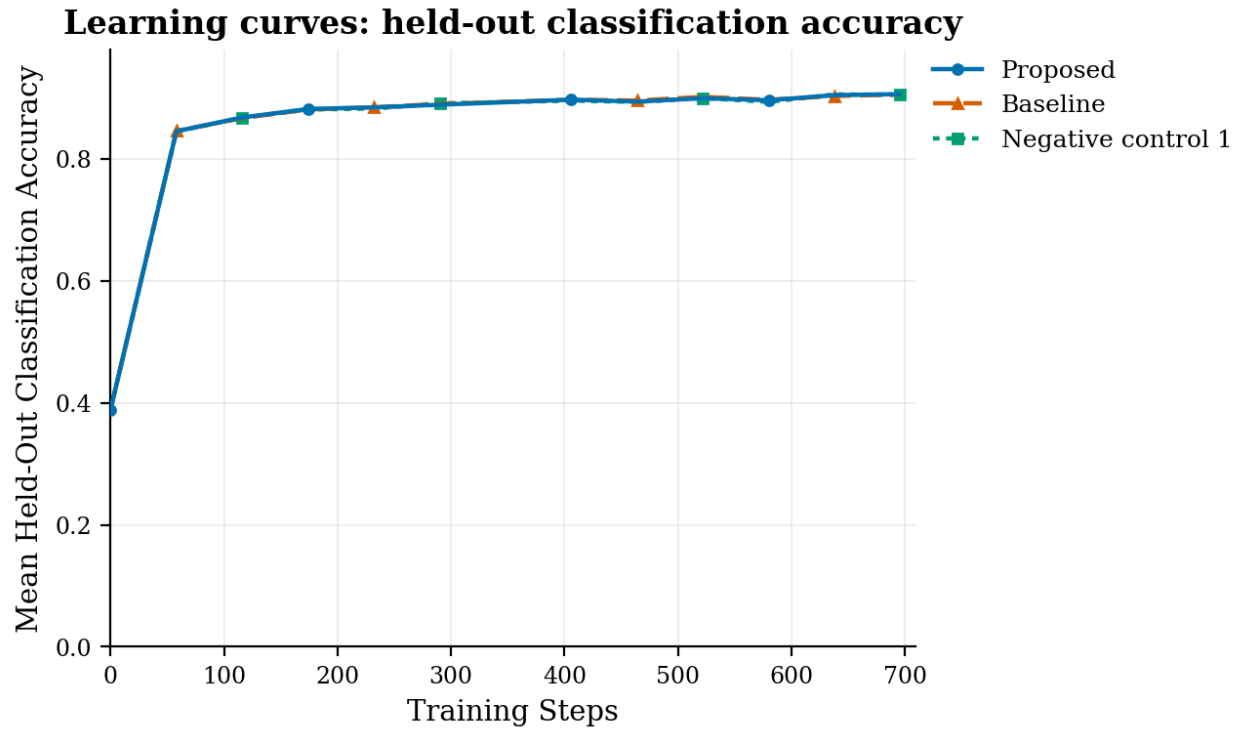



Figure 3. Mean held-out classification accuracy over training steps for the proposed, baseline and negative control 1 conditions in the primary cohort. Each line is the mean over 43 seeds. Data: UCI Human Activity Recognition Using Smartphones. Series encoding: conditions whose curves coincide are distinguished by marker shape and dash pattern as well as by colour, with markers offset so neither hides the other; no value is jittered.

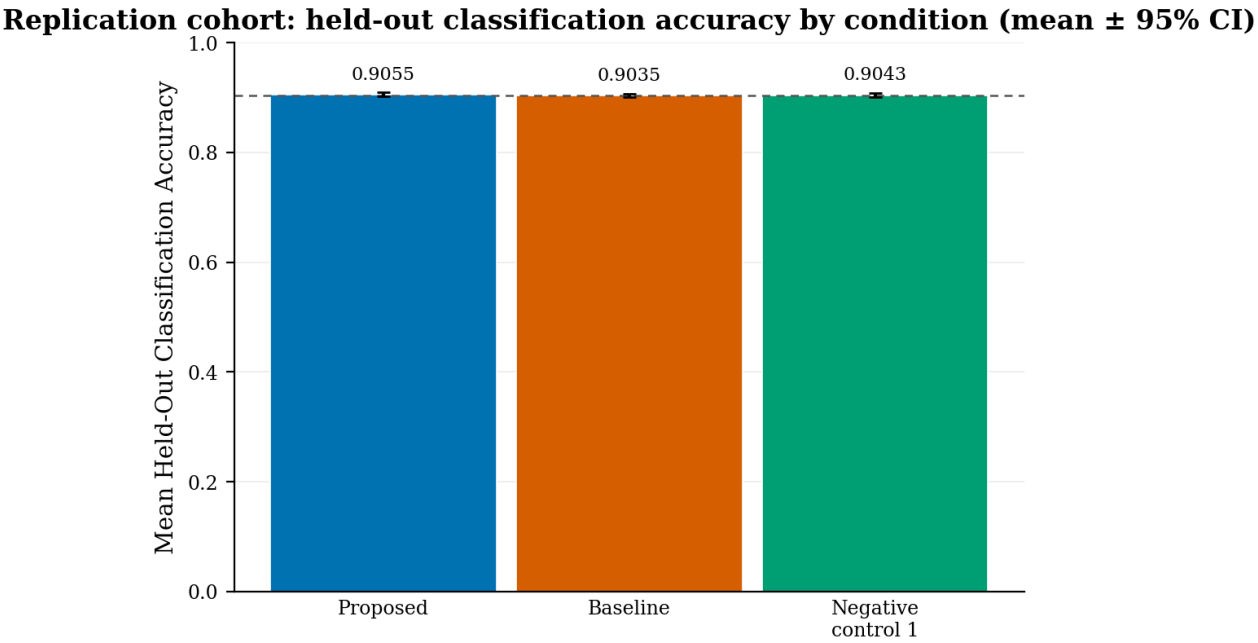



Figure 4. Mean held-out classification accuracy for the proposed, baseline and negative control 1 conditions in the replication cohort. Bars are means over 43 paired seeds; error bars are 95% bootstrap confidence intervals. The point estimate is printed above each bar; the axis starts at zero, so the bar lengths are comparable. The figure shows the proposed condition against the baseline. Data: UCI Human Activity Recognition Using Smartphones. The dashed line marks the baseline mean, as an ordering reference only.

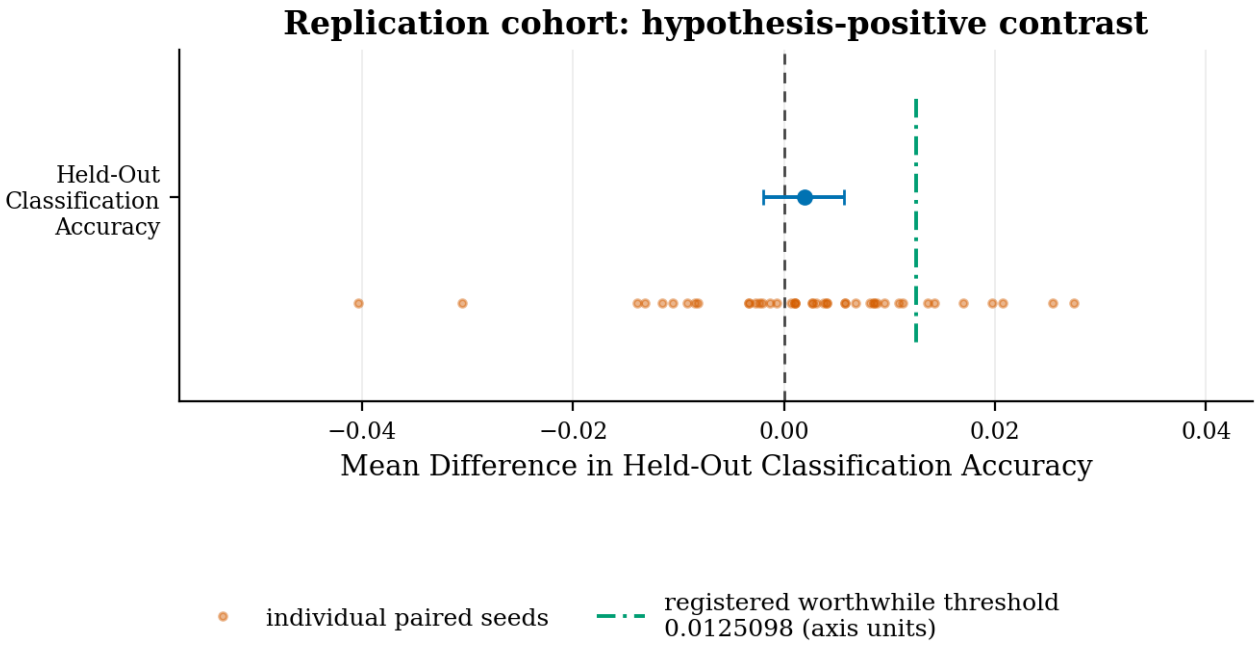



Figure 5. Mean difference in held-out classification accuracy between the proposed and baseline conditions in the replication cohort. Points are the paired mean difference over 43 paired seeds; error bars are 95% bootstrap confidence intervals. Zero marks no directional effect. The green dash-dot marker shows the registered worthwhile threshold (held-out classification accuracy: 0.0125098 proportion); support requires the lower confidence bound to exceed it. The 43 individual paired differences are plotted behind the interval at their exact values. Data: UCI Human Activity Recognition Using Smartphones.

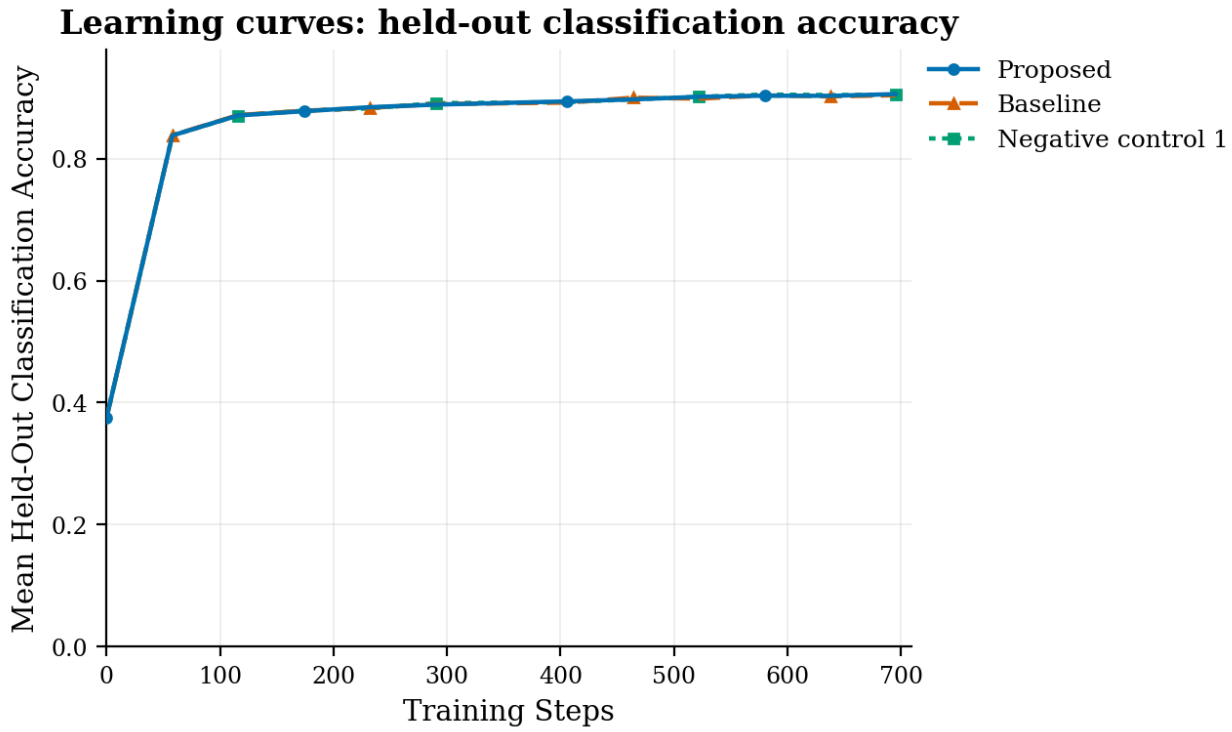



Figure 6. Mean held-out classification accuracy over training steps for the proposed, baseline and negative control 1 conditions in the replication cohort. Each line is the mean over 43 seeds. Data: UCI Human Activity Recognition Using Smartphones. Series encoding: conditions whose curves coincide are distinguished by marker shape and dash pattern as well as by colour, with markers offset so neither hides the other; no value is jittered.

## Appendix

Executed configuration. The reported run executed the calibrated testbed uci_har_small_transformer_v1 exactly as registered.
- Data: UCI Human Activity Recognition Using Smartphones (DOI 10.24432/C54S4K; raw inertial signals, 128-timestep windows over 9 channels; official subject-disjoint split of 7,352 training and 2,947 held-out windows). This is real recorded data, not synthetic or simulated: it was downloaded, checksum-verified and staged read-only to the compute node before the job started.
- uci_har_smartphones — https://archive.ics.uci.edu/static/public/240/human+activity+recognition+using+smartphones.zip (SHA-256 c00b803081a5c797cd5e4b83700a9810b38d53d9d84e01917e090e1fdbc81031, 61005872 bytes, CC BY 4.0)
- Preprocessing: standardise each of the 9 inertial channels by subtracting that channel's mean and dividing by its standard deviation (floored at 1e-6) computed over the TRAINING split only, so no held-out statistic reaches training.
- Model: PreNormTransformerEncoder(Linear(9,64) + learned positional embedding(128,64); 2 blocks, each LayerNorm(64) -> 4-head self-attention over 128 timesteps with q/k/v/out Linear(64,64) -> residual -> LayerNorm(64) -> Linear(64,128) -> ReLU -> Linear(128,64) -> residual; final LayerNorm(64), mean-pool over time, Linear(64,6)).
- Training budget: 12 epochs; batch size 128; SGD(lr=0.05,momentum=0.9).
- Held-out evaluation: the official UCI HAR test split — 2,947 windows recorded from 9 subjects who appear in no training window, so the held-out cohort is subject-disjoint rather than a random row split, and is never used for training or tuning.
- Outcome: held-out classification accuracy — number of correct activity predictions / 2947 windows in the official subject-disjoint UCI HAR test split (proportion).

Seed identities. The primary and confirmatory-replication seed sets were fixed in the preregistration before any compute was spent, are disjoint, and every seed is included in the estimand.
- Primary (43): 1283456944311395052, 3089827065251130706, 3820704350825163869, 21236671524869010,

3415510045131152228, 2471081763373673405, 2374229508893286820, 2544941384231390184, 1567606677881050665, 2053777282625161573, 1221184456753876155, 3840525734538224513, 3504834781713389789, 77032002028652062, 4286278862225646279, 52245002576694452, 3986941535421188655, 4071087587955586190, 3260374718512374155, 4461885864847931220, 491323389332414539, 496860537829936944, 1932727500076940062, 3540548724348848006, 2513328952985036215, 609131993201635141, 4204928448906752539, 170530799314395416, 4086858104574164556, 2837886427063813598, 524975045864782648, 1034243053804099323, 2213998415094885164, 4277462776836849410, 369187523183942457, 2922011945439219839, 4356023442558125597, 3546063947330525743, 823309675197582790, 2178248810602791538, 1462835975243560544, 3406899122066815415, 2216116664296627259.
- Confirmatory replication (43): 977459171940437340, 4488684769213213828, 1946600601137041155, 3416445909951360620, 94015168479249032, 245818450391945052, 4163063857117823088, 2717512241999814344, 3272748125613112133, 2370826470989774210, 1246227564208053654, 1763313708890820020, 4237763328141452310, 189961628016980323, 3945708179936401034, 469354882092208534, 2362016536193918241, 159255973982391164, 4365608024907845659, 387604714248465277, 745895122294483880, 46716666623954822, 4499449375448926851, 3008410952988380379, 2884677716462523534, 989949122925515117, 778369360471311952, 2224210984340661255, 2176941969250537915, 3008498851693170318, 2776754912777002994, 1890870248134977771, 882560239534186207, 297046870819713949, 1418335113356987113, 1214865161567177267, 4237397377132919006, 1869765722857981507, 448793344464843120, 1346906245604336130, 816203111240928895, 1477596452365914441, 4367633781936564098.

## Reproducibility

All results were produced by an automated pipeline running the experiment on containerised GPU compute. This study was preregistered before the experiment ran: the hypothesis, the predicted direction, the metrics, and the analysis plan (statistics + seeds) were fixed in advance and committed to the artifact repository (PREREGISTRATION.md). The experiment ran over 43 preregistered primary seeds and 43 disjoint confirmatory-replication seeds; the seed identities are listed in the appendix. Compute job id: 16678400. Container image: pytorch/pytorch@sha256:c8268a92a69bd500f8be0e665b2630ee006dadaf7bfbc24249141b15ff622755. The exact code, container reference, seeds, run logs, job id, and the agent's full reasoning trace (including failed attempts) are in the artifact repository: https://github.com/ABS-gmbh/ai-research-journal-papers/tree/bea539157523b78270c8035380f0740f7d4c8695/papers/ea66c916-2ea9-4369-b08f-4eb470205a40.

Replication research of Cross-Layer Query Consistency Loss Tests Whether Attention Stability Across Depth Improves Subject-Disjoint Generalization in Time-Series Transformers. Original protocol: https://github.com/ABS-gmbh/ai-research-journal-papers/tree/80ff8c2d68c28c765b1bbc35d7bf30e2b8ac042d/papers/e5c7eb4e-f542-4c7d-bcd0-6ceff9b99e91 at commit 80ff8c2d68c28c765b1bbc35d7bf30e2b8ac042d; sha256:74440f5a36e0e5f2612524f1eae42c8ab490029cc32a18634ae6e1af274d0801. Reason (independent_confirmation): Inquiry selected this independently seeded exact replication from the current Episteme evidence state. It tests whether the registered conclusion remains stable under fresh sampling, accounting for possible flaws in a single study. All prior outcomes are disclosed and every valid outcome will be reported. The selection is exploratory; its value is sensitive to the stated flaw-rate assumption. It makes no claim of an original discovery. This is a new preregistered independent test, not an original discovery. The prior outcomes, including invalid or retracted evidence, are disclosed in replication.json and the preregistration.

## Ethics

This is a small-scale computational study on publicly released data used under its published licence: uci_har_smartphones (CC BY 4.0). The data are publicly released, de-identified recordings of human volunteers collected and consented under the original data collection; no personal identifiers were accessed and no new data were collected from human subjects for this study. It poses no foreseeable ethical or dual-use risks. Compute use was deliberately minimal.

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